18 April 1988

MILITARY SPECIFICATION

RESISTOR, FIXED, METAL ELEMENT (POWER TYPE),

STYLE RLV20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a style RLV20 metal element, power type, fixed resistor.

1.2 Military part number. The resistor covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-R-49465	-	Resistors, Fixed, Metal Element (Power Type), (Very Low Resistance Values), General Specification For.
MIL-R-39032	-	Resistors, Packaging of.

(Copies of the specifications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-49465, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Characteristic. This style resistor is available in characteristic L as described in MIL-R-49465.

3.4 Tolerance. This style resistor is available in tolerances J and K (±5 percent and ±10 percent respectively).

3.5 Terminal lead types. This style resistor is a four-leaded device with solderable terminals (see figure 1).

3.6 Weight. The weight shall be as specified on figure 1.

3.7 Power rating. The power rating shall be as shown in table I.

3.8 Maximum overload current. The maximum overload current shall be as specified in table I.

3.9 Resistance value range. The minimum and maximum resistance values shall be as specified in table I.

3.10 Temperature coefficient. The temperature coefficient of resistance shall be as specified in table II.

3.11 Marking. The resistor shall be marked in accordance with MIL-R-49465.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-49465.

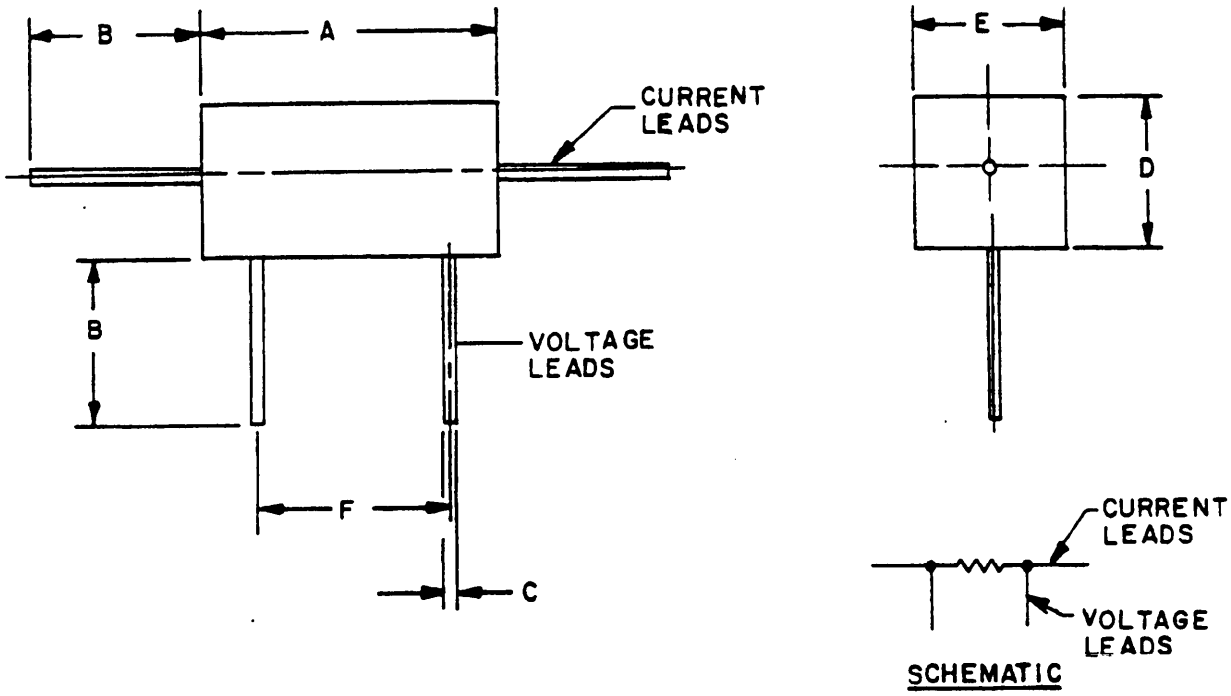
4.2 Visual and mechanical inspection. Visual and mechanical inspection shall be in accordance with MIL-R-49465.

4.3 Terminal strength. The applied force shall be five pounds.

TABLE I. Electrical characteristics.

Standard style	Rated wattage at +25°C (watts)	Minimum resistance (ohms)	Maximum resistance (ohms)	Maximum overload current (amperes)
RLV20	3.0	.010	0.10	32.0

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Standard style	Maximum weight lbs (grams)	Dimensions					
		A ±.031 (±.787)	B Min.	C ±.002 (±.050)	D ±.031 (±.787)	E ±.031 (±.787)	F ±.060 (±.152)
RLV20	.011 (5.0)	.875 (22.23)	1.00 (25.4)	.036 (.914)	.312 (7.92)	.312 (7.92)	.562

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. Lead length shall be 1 inch minimum.
4. For resistance measurements the point of contact to the leads shall be .375 (9.52 mm) ±.062 (1.57 mm) from the body.
5. The picturization of the styles above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.

FIGURE 1. Resistor, fixed, metal element.

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TABLE II. Temperature coefficient of resistance.

Standard style	.01 ohm to .0249 ohm	.025 ohm to .0499 ohm	.05 ohm to .0749 ohm	.075 ohm to .099 ohm	0.1 ohm and above
	(ppm/°C)	(ppm/°C)	(ppm/°C)	(ppm/°C)	(ppm/°C)
RLV20	±100	±100	±100	±100	±100

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

6.1 Notes. The notes specified in MIL-R-49465 are applicable to this specification.

6.2 Ordering data. Ordering data shall be in accordance with MIL-R-49465.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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